

Store Your Battery Right

Stored batteries can lose 2% to 5% of capacity every month at 25° C. Higher the temperature higher the loss.

Give stored batteries a maintenance charge once in 3 months.

www.relicellbattery.com



VRLA and Tubular batteries

VRLA Batteries - 12V 7AH to 250 AH
Tubular Batteries - 12V 40AH to 210 AH

Part of a Rs.1000 crore IT industry group
Tested and certified by Central Power Research Institute, Bangalore (CPRI) and Electronic Test and Development Centre (STQC Directorate, Ministry of Information Technology, Govt. of India). **Download Test Certificates at** www.relicellbattery.com

UL Certified

VRLA Batteries approved and registered in DGS&D for government purchases
47 Service Centres across India

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E-Waste Management

We thank EFY and M. Balaji Prasad, the author, for publishing the article 'Creating and Sustaining an Effective e-Waste Management System' in June issue. The research work and the article are really appreciable. We expect more information on the technology used for e-waste management.

Rajesh Patel

Globe E-Waste Management
Sagun Ind. Estate, Surat

EFY Awards

I suggest you to include 'The Best Electronics Repairing and Refurbishing Services' category in EFY Awards with the following sub-categories: telecom, consumer electronics and IT. This industry is coming up rapidly in India with global players in this field establishing base here.

Manoj Kumar Singh
Gurgaon

EFY: Thanks, your suggestion has been passed on to the EFY Awards committee for consideration.

PCB Design Software

I need free and open source software (FOSS) for PCB design. Please help.

Ameya Singh
Through e-mail

EFY: Kicad, FreePCB, PCB Editor and TinyCAD are some of the known FOSS PCB design software. You can download them from their respective websites.

Low-power Products

In the 'Designing Low-power Products Using MSP430 Microcontroller' construction project published in January and February 2009 issues, the Flash emulation tool (FET) supports MSP430F1491 and MSP430F1611 MCU chips on the code composer essential (CCE) platform perfectly. But in the case of MSP430F2132 MCU, FET detects it only for a short period of time. I have confirmed that all the settings in CCE debuggers are correct. In the article, you have mentioned that FET will support all MSP430 family

MCUs. Please help.

Vinod Kr Singh, Engineer (R&D)
Karnataka Hybrid Micro Devices Ltd
Bangalore

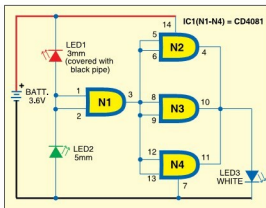
The author **Harinath Renukumrthy** replies:

Yes, the FET programmer should support all MSP430 devices. First, check whether the target board connections containing F2132, especially JTAG ports (TDO, TCLK, TMS, TCK, TEST), DVSS, DVCC, RST and XIN/XOUT pins, are correctly connected. Second, verify the current consumption of the target board. It should be less than 2 mA, unless extra power is supplied through the USB cable.

I recommend you to download the latest version of CCE, called CCS, from Texas Instruments' webpage <http://focus.ti.com/mcu/docs/mcuproductmstools.w.jsp?sectionId=95&tabId=1203&familyId=342&toolTypeId=1#IDE>

Emergency Photo Lamp

In the 'Emergency Photo Lamp' circuit published in May issue, photo-transistor T1 and bipolar transistor T2 may be replaced with two transparent LEDs as shown in the figure here. Typically, an LED under normal lighting conditions



Modified circuit for emergency photo lamp

can pass a current of the order of nano-amperes (nA). When light falls on the LED, the current increases from tens of nA to hundreds of nA depending on the intensity of light. Thus the white LED glows when light does not fall on LED1 and LED2.

Raju R. Baddi
The author